

MAËL CHAUMETTE

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🐙 GitHub 🪄 GitLab 🌐 LinkedIn

EDUCATION

National School for Statistic and Data Analysis - ENSAI

2021 - 2024

Master's degree in Data Science and Data engineering

Rennes, France

- Relevant courses: Deep Learning, Machine Learning, Numerical optimization, Multivariate statistical analysis, Advanced inferential statistics, Probability, Measure theory

University of Rennes

2021 - 2022

Bachelor of Science in Mathematics with high honors

Rennes, France

- Dual curriculum in conjunction with ENSAI studies
- Relevant courses: Complex analysis, Differential equations, Numerical optimization

EXPERIENCE

ENS Lyon / Inria - OCKHAM team

Nov. 2024 - Present

Ph.D supervised by R. Gribonval and E. Riccietti

Lyon, France

- “Invariance, parsimony and quantification for frugal and controlled large-scale learning”
- Keywords: deep learning, quantization, optimization, sparsity
- Publications [3, 4, 1, 2]

ENS Lyon / Inria - OCKHAM team

Apr. 2024 - Sep. 2024

Research internship (M2) supervised by R. Gribonval and M. Massias

Lyon, France

- Numerical optimization by studying the spectrum of the objective function's Hessian
- Tools & technologies used: Python, Numerical optimization, Random matrix and Measure theory

National School of Public Health - EHESP

Jun. 2023 - Aug. 2023

Machine learning internship to predict hospital readmissions

Rennes, France

- Utilisation of Spark to handle massive data
- Tools & technologies used: Python, Scala, Spark, Random Forest, Logistic regression

PUBLICATIONS

Preprint

[1] M. Chaumette, R. Gribonval, E. Riccietti. On the Real Part of Butterfly Factorizations, with Application to Fast Transforms. Submitted at *Signal Processing Letters*, 2026

[2] M. Chaumette, R. Gribonval, E. Riccietti. CROQuant: Complex Rank-One Quantization Algorithm, with Application to Butterfly Factorizations. Submitted at *ACM Transactions on Mathematical Software*, 2026

National Conference

[3] G. Lauga, M. Chaumette, E. Desainte-Maréville, E. Lassalle, A. Lebeurrer. A multilevel approach to accelerate the training of Transformers. *GRETSI*, 2025

[4] M. Chaumette, R. Gribonval, E. Riccietti. CROQuant: Complex Rank-One Quantization Algorithm. *GRETSI*, 2025

TALKS

Quantization of rank- r matrices with application to Discrete Cosine Transform and image compression. *STOP*, Oct 2026

CROQuant: Complex Rank-One Quantization Algorithm, with Application to Butterfly Factorizations. Poster at *SMAI-MODE*, Mar 2026

CROQuant: Complex Rank-One Quantization Algorithm. *RAIM*, Nov 2025

CROQuant: Complex Rank-One Quantization Algorithm. *GRETSI*, Sep 2025

TEACHING

2025 - 2026 (32h)	ENS de Lyon: Teacher Assistant of Mathurin Massias, <i>Machine Learning (L3)</i>
2024 - 2025 (32h)	ENS de Lyon: Teacher Assistant of Stephan Thomassé, <i>Algorithms 1 (L3)</i>

TECHNICAL SKILLS AND INTERESTS

IT Skills	Python (Pytorch), R, Java, Spark, Git
Other skills	Driven licence, Team First AID 2
Languages	French (Native speaker), English (845 TOEIC)
Hobbies	Swimming, Travelling, cinema